

Pharmacotherapy of hypertension

Introduction

Hypertension

- ▶ Defined as persistently elevated arterial blood pressure: $\text{SBP} \geq 130 \text{ mmHg}$ and/or $\text{DBP} \geq 80 \text{ mmHg}$
- ▶ Is one of the most significant risk factor for cardiovascular diseases
 - ▶ Starting from 115/75 mmHg, the CVD doubles with every 20/10 mmHg increase in BP.

Epidemiology

- ▶ HTN is major cause of premature death worldwide
- ▶ Globally doubled from 650 million to 1.3 Billion from 1990 to 2019
- ▶ From this 2/3rd are living in LMIC
- ▶ 46% of adults are unaware that they have HTN
- ▶ Less than 42% with HTN are Diagnosed & treated
- ▶ Only 21% with HTN have it under control
- ▶ Prevalence in Ethiopia: between 20% and 30%

(Reference WHO, 2023)

Epidemiology

▶ Sex

- Age < 45  Men>>>>women
- Age 45 – 64  similar for both sex
- Age > 64  women >>>>men

▶ BP values increase with age

- ▶ hypertension - very common in the elderly

Risk factors

- Age (greater than or equal to 55 for men, greater than or equal to 65 for women)
- Diabetes mellitus
- Dyslipidemia
- Microalbuminuria (A2=Albumin loss >30mg/day)
- Family history of premature CV disease
- Obesity
- Physical inactivity
- Excessive intake of salt, alcohol or caffeine
- Tobacco use
- Stress

Etiology

- ▶ *Essential or primary hypertension*
 - ▶ 90-95% of patients
 - ▶ Results from **unknown** pathophysiological etiology
 - ▶ Monogenic and polygenic forms of BP dysregulation
 - ▶ genetic mutations altering
 - urinary kallikrein excretion
 - nitric oxide release, and
 - secretion of aldosterone, other adrenal steroids, and angiotensinogen
 - Sodium balance

Etiology

- ▶ *Secondary hypertension*

- ▶ 5-10% of patients

- ▶ Results from a specific cause (a comorbid disease or drug or other product)

- a. Renal disorders [Reno-vascular dz, renal parenchymal dz, renin-secreting tumors]*

- b. Endocrine disorders [Cushing's syndrome(Cortisol), Pheochromocytoma]*

- c. Vascular disorders [Coarctation of aorta, Rigidity of aorta]*

- d. Pregnancy-induced HTN*

- e. Neurological disorders*

- f. HTN after cardiac surgery*

- g. Drugs and alcohol*

- 5% of OCP users develop HTN within 5yr

Etiology

- ▶ Drugs or other products
 - Amphetamines, Corticosteroids
 - Calcineurin inhibitors: cyclosporine, tacrolimus
 - Decongestants: pseudoephedrine, phenylephrine
 - Ergot alkaloids: ergonovine, methysergide
 - ESA: erythropoetin, darbopoetin
 - Estrogen-containing oral contraceptives
 - NSAIDs
 - Cocaine & nicotine and withdrawal
 - Abrupt discontinuation of BB or alpha 2 agonists
 - MAO inhibitor with tryamine containing foods
 - Sodium, Ethanol

Pathophysiology

- ▶ Arterial BP is the pressure in the arterial wall
 - ▶ Systolic blood pressure (SBP)
 - represents the **peak value**
 - achieved during cardiac contraction
 - ▶ Diastolic blood pressure (DBP)
 - represents the **nadir** value
 - achieved after contraction when the cardiac chambers are filling

Pathophysiology

- ▶ $BP = CO \times TPR$
 - ▶ $CO = HR \times SV$
- ▶ **Pulse pressure** = $SBP - DBP$
 - ▶ A measure of arterial wall tension (arterial stiffness)
 - ▶ Reflect extent of atherosclerotic disease in the elderly
- ▶

Pathophysiology

▶ **MAP** = $\left(\text{SBP} \times \frac{1}{3}\right) + \left(\text{DBP} \times \frac{2}{3}\right)$

- ▶ The average pressure throughout the cardiac cycle
- ▶ Used in hypertensive emergency.
- ▶ CO is the major determinant of SBP
- ▶ TPR is the major determinant of DBP

Pathophysiology

▶ BP regulations

A. Humoral Factors

- ▶ RAAS: regulates systemic and renal blood flow
- ▶ Natriuretic hormone
- ▶ Insulin resistance and hyperinsulinemia

B. Neural Factors

- ▶ SNS: controls α - and β -adrenergic receptors

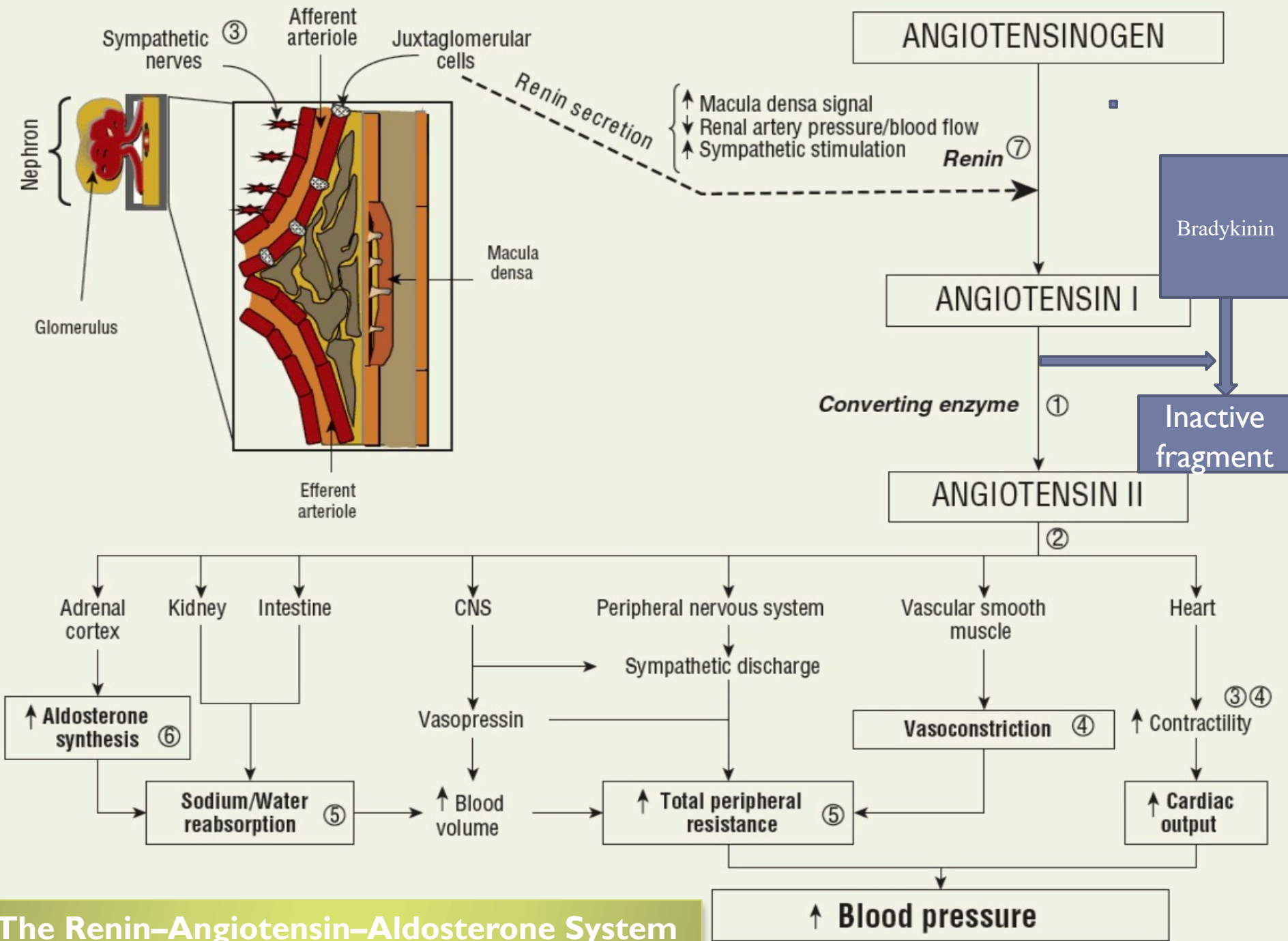
C. Peripheral auto-regulatory components

D. Vascular endothelial mechanisms

- ▶ Regulates release of nitric oxide [NO], bradykinin, prostacyclin, endothelin

E. Electrolytes

Disturbances in sodium, calcium



Classification

- ▶ The ACC/AHA classification of BP in adults (age ≥ 18)
 - ▶ based on the average of 2 or more properly measured BP values from 2 or more clinical encounters

BP Category	SBP		DBP
Normal	<120 mm Hg	and	<80 mm Hg
Elevated	120–129 mm Hg	and	<80 mm Hg
Hypertension			
Stage 1	130–139 mm Hg	or	80–89 mm Hg
Stage 2	≥ 140 mm Hg	or	≥ 90 mm Hg

- ▶ If systolic and diastolic BP values yield different classifications, the highest category is used for the purpose of determining a classification.

Classification

- ▶ *Isolated systolic hypertension*

- when SBP values are elevated (≥ 130 mm Hg) and DBP values are not (< 80 mm Hg)
- Common among the elderly

Classification

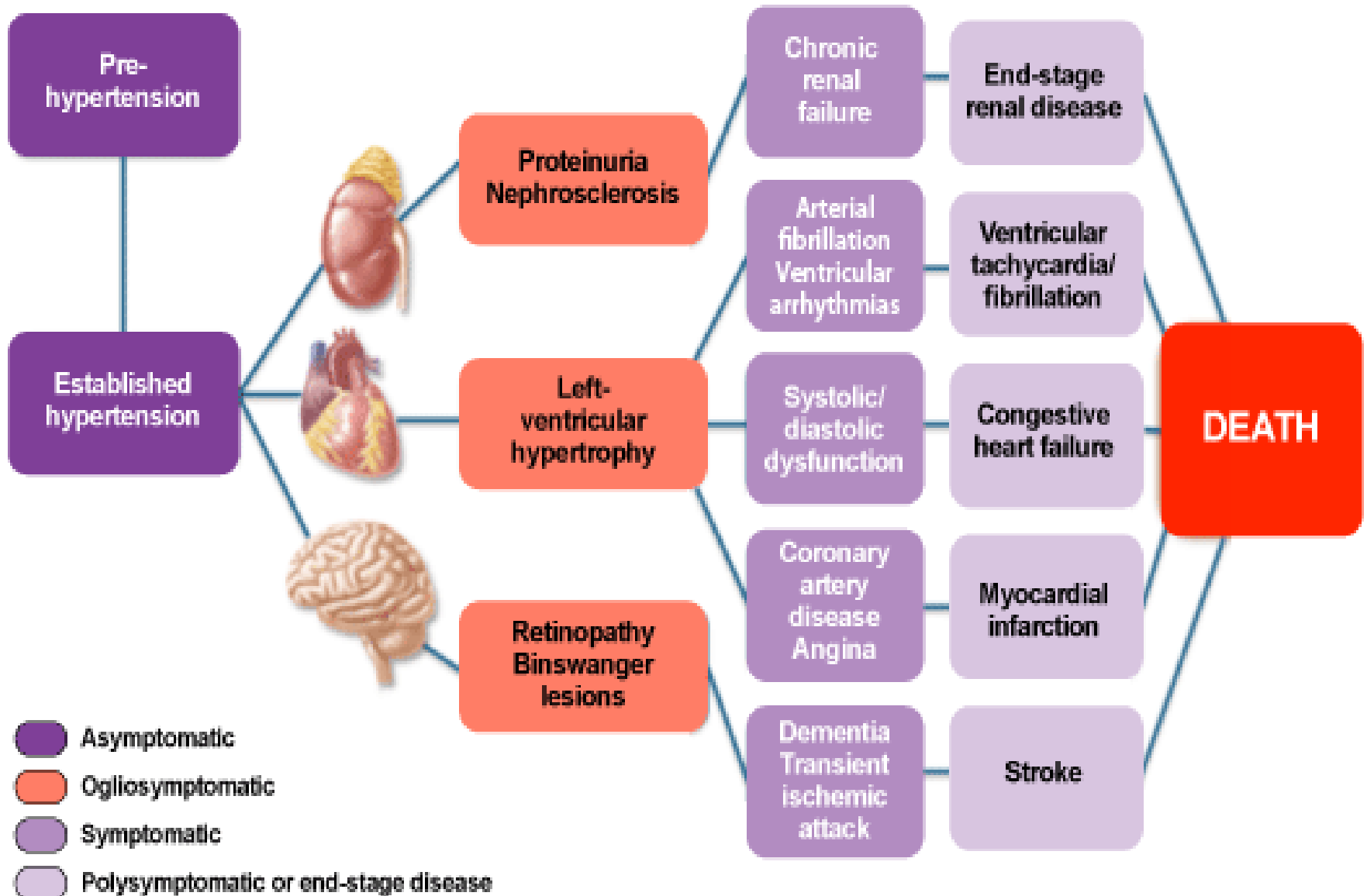
▶ *Hypertensive crises*

- Clinical situations where BP values are elevated
- Greater than 180/120 mm Hg

▶ Categorized as either *hypertensive emergency* or *hypertensive urgency*

- **Hypertensive emergencies:** extreme elevations in BP accompanied by acute or progressing target-organ damage
- **Hypertensive urgencies:** high elevations in BP without acute or progressing target-organ injury

Complication of Hypertension



Clinical presentation

- ▶ *“silent killer”*
 - the patient may appear healthy or may have the presence of additional CV risk factors
- ▶ The patient may complain of
 - Headache
 - Dizziness
 - Nervousness
 - Flushed face
 - Lack of sleep

DIAGNOSIS

- ▶ Frequently, the only sign of HTN on **physical examination** is elevated BP.
- ▶ The diagnosis of HTN should be based on the average of two or more readings taken at each of two or more clinical encounters.
- ▶ As HTN progresses, signs of end-organ damage begin to appear, chiefly related to pathologic changes in the **eye, brain, heart, kidneys, and peripheral blood vessels**

Diagnosis...

- ▶ **Cardiovascular effects:** elevated BP, difficulty in breathing, fluid retention in extremities, heart murmurs
- ▶ **CNS effects:** headaches, drowsiness, anxiety and mental impairment
- ▶ **Renal:** polyuria, nocturia, elevated serum creatinine
- ▶ **Ocular changes:** Fundoscopic changes, arteriolar narrowing, and retinal hemorrhages

Case study

- ▶ A.M. is a 37 years old female patient who was diagnosed with hypertension with a BP of 135/95mm Hg and was on life style modifications but failed to control her BP. The patient was diagnosed based on the average of two BP measurements on two visits. A.M. Is a mother of three children who kept her busy in life. She lives with her husband who works as a daily laborer to support his family. The patient doesn't smoke and exercise. She feeds a "normal diet" which is high in sodium. The patient is not pregnant and no near future plan to become pregnant and have no other comorbid condition.

Case study

- a. What is the stage of A.M.'s hypertension?
- b. What are the goals of therapy for the patient.
- c. What are the non-pharmacologic therapy options for the patient.
- d. Design drug therapy regimen for the patient.
- e. What monitoring parameters should be followed? (effectiveness Vs Safety)

Treatment

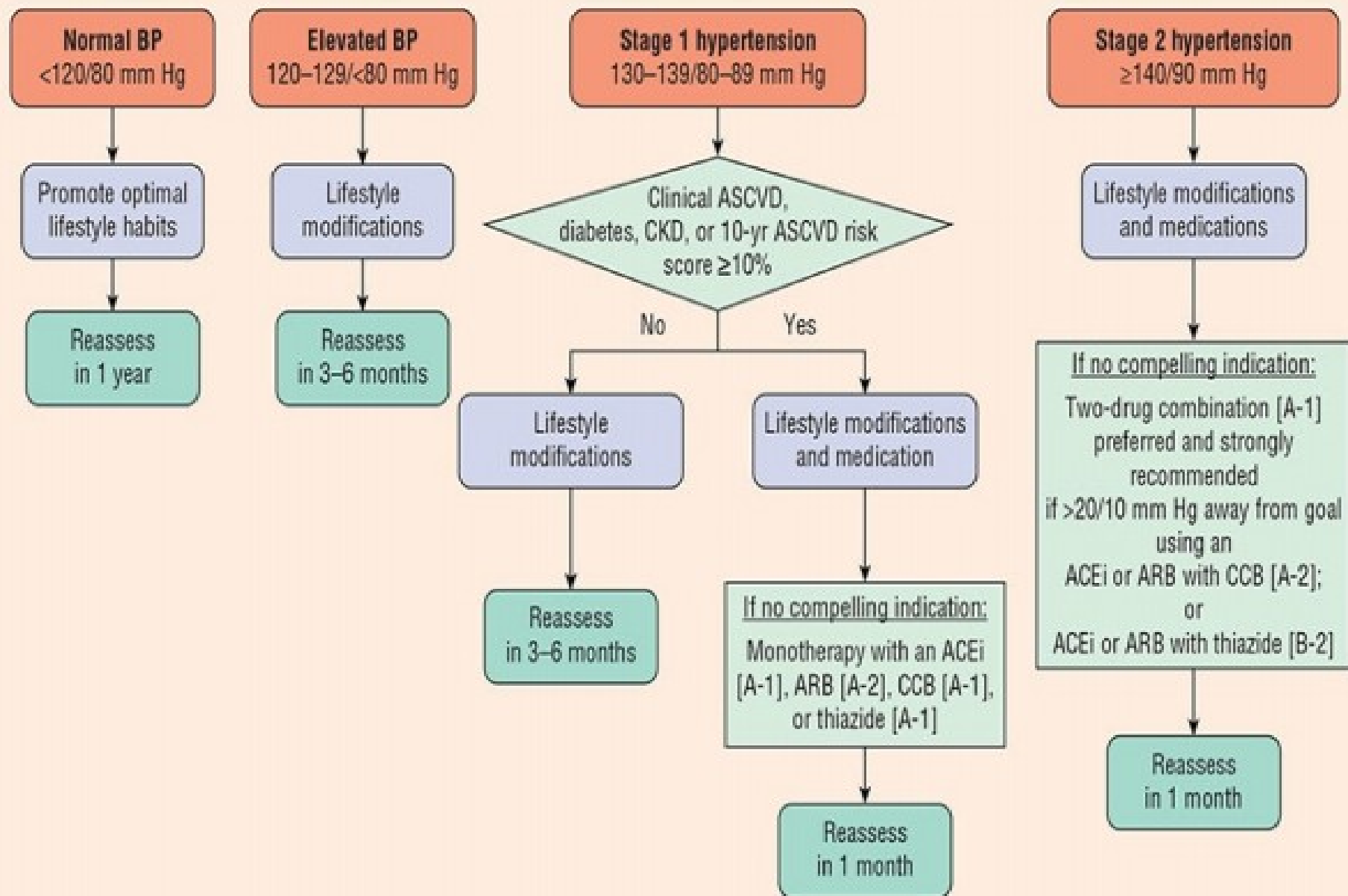
Goal of therapy

- To reduce associated morbidity and mortality
- To reduce hypertension-associated complications
- To reduce side effects of antihypertensive medications
- To reduce CV risk
- To maintain goal BP
 - BP target < 130/80 mm Hg (ACC/AHA)

Treatment

- Lifestyle modifications and drug therapy essential for most patients
- Lifestyle modification alone is appropriate therapy for most patients with
 - elevated BP
 - Stage I hypertension and 10-year ASCVD event < 10%
- For patients with hypertension and additional CV risk factors or hypertension-associated target organ damage
 - Lifestyle modifications + drug therapy

Hypertension Treatment Algorithm



Effect of life style recommendations

Modifications	Recommendation	SBP reduction
Weight loss	Maintain normal body weight (body mass index, 18.5–24.9 kg/m ²)	5–20 per 10-kg weight loss
dietary patterns	Consume a diet rich in fruits, vegetables, and low-fat dairy products with a reduced content of saturated and total fat	8–14
Reduced salt intake	Reduce daily dietary sodium intake as much as possible, ideally to 65 mmol/day (1.5 g/day sodium, or 3.8 g/day sodium chloride)	2–8
Physical activity	Regular aerobic physical activity (at least 30 minutes/day, most days of the week)	4–9
Moderation of alcohol intake	Limit consumption to ≤ 2 drink equivalents per day in men and ≤ 1 drink equivalent per day in women and lighter weight persons	2–4

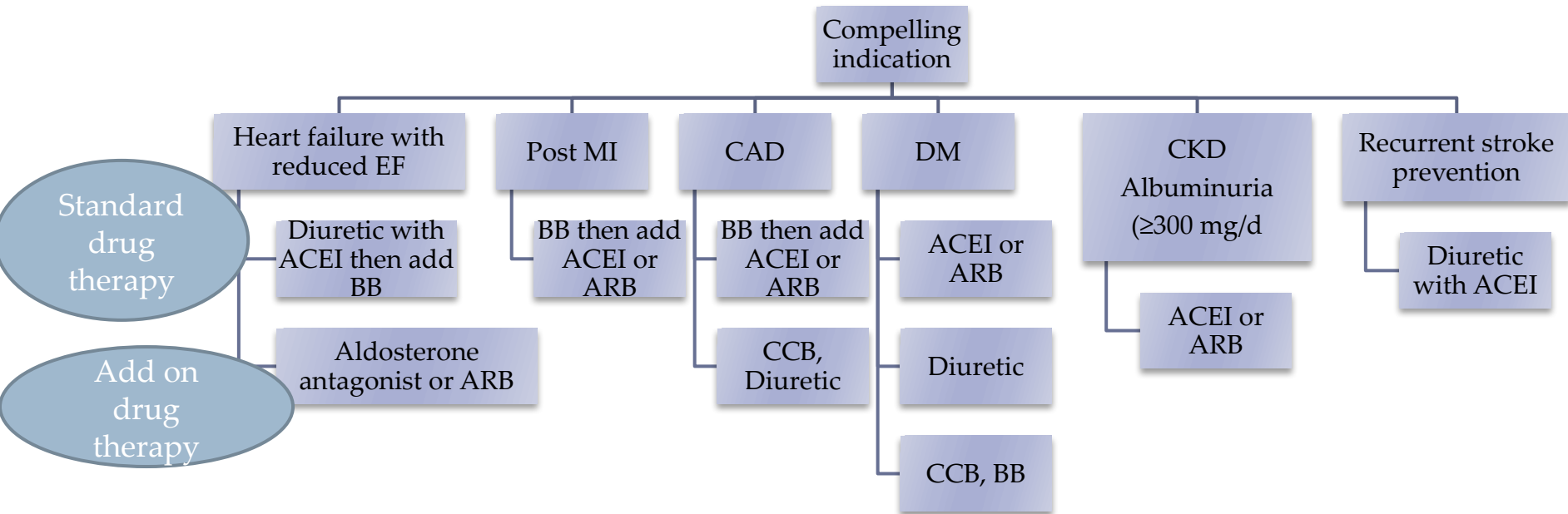
Drug therapy

- ▶ The choice of initial drug therapy depends on
 - ▶ The degree of BP elevation and
 - ▶ presence of compelling indications
- ▶ Patients with stage I hypertension
 - ▶ initial treatment with **one first-line antihypertensive drug**
- ▶ Patients with stage II hypertension
 - ▶ **Combination drug therapy** with two first-line antihypertensive drugs from different drug class

Drug therapy

- ▶ first-line antihypertensive drugs *for patients without a compelling indication for a specific drug class*
 - ACEIs
 - ARBs
 - CCBs
 - Thiazide type diuretics
- ▶ **Alternative** antihypertensive drugs
 - BBs
 - Central Alpha 2 agonists
 - Direct vasodilators & peripheral adrenergic antagonists
 - Renin Inhibitors

Drug therapy based on compelling indication



Combination Therapy in HTN

- ▶ If monotherapy fails the clinician can
 - Increase the dosage of the first agent (can increase S/E)
 - Replace the first agent (may be suboptimal if the 1st agent is working but BP is not controlled)
 - Or add other agents
- ▶ If a diuretic is not used initially, it should be added to the regimen
- ▶ If the patient is stage II HTN, Combination therapy is the choice of treatment

Recommendations for Combination Therapy

Preferred

- ACE inhibitor/CCB
- ARB/CCB
- ACE inhibitor/diuretic
- ARB/diuretic

Acceptable

- β -Blocker/diuretic
- CCB (dihydropyridine)/ β -blocker
- CCB/diuretic
- Renin inhibitor/diuretic
- Thiazide diuretics/potassium-sparing diuretics

Less Effective

- ACE inhibitor/ β -blocker
- ARB/ β -blocker
- CCB (nondihydropyridine)/ β -blocker
- Centrally acting agent/ β -blocker

ACE Inhibitors

- Captopril: 12.5-150 mg, BID or TID
- Enalapril: 5-40 mg, QD or BID
- Fosinopril: 10-40 mg, QD
- Lisinopril: 10-40 mg, QD
- Benazepril: 10-40 mg, QD or BID
- Quinapril: 10-80 mg, QD or BID
- Ramipril: 2.5-10 mg, QD or BID

ACE Inhibitors

- ▶ Inhibits the conversion of angiotensin I to angiotensin II
 - ↓ rate of deactivation of bradykinin (a potent vasodilator)
- ▶ Does not cause fluid retention, thus reducing diuretic requirements
- ▶ Reduce after load and preload, thereby reducing oxygen demand and myocardial wall stress
- ▶ Dilate systemic arterial resistance vessels, thus reducing BP

ACE Inhibitors

- ▶ Standard of care in patients with **MI and CHF**
- ▶ Preferred antihypertensive agent in patients with
 - Diabetes
 - CKD
- ▶ Administered once or twice daily (except captopril)
- ▶ Angioedema is a life threatening side effect that can occur anytime during treatment



ACE Inhibitors

- ▶ Avoid use in
 - ▶ Dialysis patient
 - ▶ Concomitantly with potassium sparing diuretics, ARBs, direct renin inhibitors, potassium supplements
 - ↑ potassium levels
- Contraindicated in bilateral renal artery stenosis or unilateral renal artery stenosis in patients with one functioning kidney

ACE Inhibitors

- ▶ May induce hypotension in
 - Sodium or volume depleted patients
 - Heart failure exacerbation
 - Very elderly, or
 - Concurrent vasodilators or diuretics
 - Start at half the normal starting dose
 - Titrate over time

ACE Inhibitors

- ▶ Side effects include:
 - **Dry cough: 20%** of treated patients
 - Angioedema: <1% of the population
 - **lip and tongue swelling**
 - **difficulty breathing**
 - Hyperkalemia
 - Rash
 - loss of taste and leukopenia
 - Elevation in SCr- if absolute increase > 1 mg/dl from baseline reduce dose or stop therapy
- ▶ Should not be used in pregnancy (Category D)
- ▶ Adjust doses with renal insufficiency (except fosinopril-hepatic elimination)

Angiotensin Receptor Antagonist

- Candesartan : 8-32 mg, QD or BID
- Irbesartan: 150-300 mg QD
- Losartan: 50-100 mg QD or BID
- Valsartan: 80-320 mg QD
- Eprosartan: 600-800 mg QD or BID
- Olmesartan: 20-40 mg QD
- Telmisartan: 20-80 mg QD

Angiotensin Receptor Antagonists

- ▶ Angiotensin II generated by two enzymatic pathways:
 - The RAAS: which involves ACE
 - An alternative pathway: uses other enzymes such as chymase (aka “tissue ACE”)
- ▶ ARBs directly block the AT₁ receptor that mediates the known effects of angiotensin II
 - Vasoconstriction
 - Aldosterone release
 - Sympathetic activation
 - Antidiuretic hormone release, and
 - Constriction of the efferent arterioles of the glomerulus

Angiotensin Receptor Antagonists

- ▶ Do not block the AT₂ receptor
 - ▶ Beneficial effect of AT II intact: vasodilation, tissue repair, and inhibition of excess cell growth
- ▶ Do not block the breakdown of bradykinin
 - ▶ beneficial effects of bradykinin
 - ▶ vasodilation, regression of myocyte hypertrophy and fibrosis, and increased levels of tissue plasminogen activator not present with ARB therapy

Angiotensin Receptor Antagonists

- Doesn't cause dry cough
- May cause angioedema (but rare)
- Efficacy similar to other antihypertensives with fewer side effects
- More expensive than ACEIs and their effect on mortality risk is not well established
- Monitor for renal insufficiency and hyperkalemia

Angiotensin Receptor Antagonists

- ▶ Side effects include
 - renal insufficiency
 - hyperkalemia, and
 - orthostatic hypotension
- ▶ ARBs should not be used in pregnancy

Calcium Channel Blockers

▶ Dihydropyridine CCBs

- Amlodipine: 2.5- 10 mg, QD
- Nifedipine: 30 -90 mg, QD
- Felodipine: 5-20mg, QD
- Isradipine: 5-10 mg, BID

▶ Non Dihydropyridine CCBs

- Diltiazem (SR): 180-360 mg BID
- Verapamil (SR): 180-480 mg QD or BID

Calcium Channel Blockers

- ▶ **MOA:** block inward movement of calcium causing relaxation and dilation
- ▶ **Dihydropyridines**
 - ▶ vasodilators of peripheral and coronary arteries
- ▶ **Nondihydropyridines**
 - ▶ Reduce arteriovenous nodal conduction
 - ▶ Negative chronotropic and inotropic actions

Calcium Channel Blockers

- ▶ Dihydropyridine CCBs
 - ▶ Effective in older patients with **isolated systolic hypertension**
- ▶ Should be used with caution in heart failure patients (Non-dihydropyridine CCBs)
- ▶ Do not use Non-dihydropyridine CCBs with BBs

Calcium Channel Blockers

- ▶ Side effects include
 - Flushing
 - peripheral edema
 - Reflex tachycardia (low with amlodipine)
 - bradycardia or heart block, and
 - constipation

Thiazide Diuretics

▶ Thiazide Diuretics

- Chlorothalidone: 6.25–25 mg, QD
- Hydrochlorothiazide: 12.5–25 mg, QD

Thiazide Diuretics

- $\uparrow$ Na/water excretion, $\downarrow$ extracellular volume, $\downarrow$ CO, $\downarrow$ BP
- Most commonly used
- Proved to be effective in reducing hypertension associated target organ damage
- Thiazides most useful in the elderly
- Thiazides not effective in patients with renal failure (GFR of <30 mL/min/1.73 m²), use loop diuretics instead

Thiazide Diuretics

- ▶ Thiazides preferred in patients with osteoporosis
 - ▶ ↓ calcium excretion
- ▶ Caution in gout patients
 - ▶ ↑ uric acid conc.
- ▶ Caution in hyperlipidemia and diabetic patients
 - ▶ ↑ blood glucose and cholesterol levels

Thiazide Diuretics

▶ Side effects:

- hypokalemia/hyperuricemia/hyponatremia
- Increases cholesterol & glucose levels
- Increases uric acid, calcium, photosensitivity
- Caution should be taken with sulfur allergic patients
- May cause volume depletion

Thiazide Diuretics

- ▶ Monitor for electrolyte imbalances with diuretics
 - ▶ Monitor K, SrCr, lipids, glucose and uric acid levels
- ▶ Dose in the morning????
- ▶ Thiazide diuretics preferred for chronic therapy

Potassium sparing diuretics

- ▶ Used in combination with thiazide or loop diuretics
- ▶ Counteracts the potassium depleting properties of other diuretic agents
- ▶ Includes
 - Amiloride: 5-10 mg, QD or BID
 - Triamterine: 50-100 mg, QD
 - Spironolactone: 25-50 mg, QD or BID
 - Eplerenone: 50 -100, QD or BID
- ▶ Spironolactone and eplerenone block the aldosterone receptor which inhibits
 - ▶ sodium and water retention

Potassium sparing diuretics

- ▶ Dose in the morning and late afternoon (when twice daily)
- ▶ Used in combination with thiazide-type diuretics
 - ▶ To minimize hypokalemia
- ▶ Reserved for patients experiencing diuretic-induced hypokalemia
- ▶ Avoid potassium sparing diuretics
 - ▶ If significant renal dysfunction (estimated creatinine clearance <45 mL/min)

Potassium sparing diuretics

- ▶ Side effects
 - ▶ Hyperkalemia
 - Dose-dependent
 - Prominent in patients with CKD or in patients taking a concurrent ACEI or ARB
 - ▶ Gynecomastia- spironolactone

Beta-Adrenergic Blockers

- ▶ Atenolol: 25-100 mg QD
- ▶ Metoprolol: 50-200 mg QD
- ▶ Bisoprolol: 2.5-10 mg QD
- ▶ Nadolol: 40-120 mg QD
- ▶ Propranolol: 40-320 mg QD

Beta-Adrenergic Blockers

▶ Intrinsic Sympathomimetic Activity

- Acebutolol
- Catechol
- Penbutolol
- Pindolol

▶ Alpha-Beta Blockers

- Carvedilol
- Labetalol

▶ Cardioselective β -blockers

- atenolol, bisoprolol, metoprolol, nebivolol

▶ nonselective β -blockers

- propranolol, nadolol

Beta-Adrenergic Blockers

- 1st line agents for patients with Myocardial Infraction (MI) & HTN
- Considered as initial therapy for patients with HTN & angina pectoris
- Also used for arrhythmias (PSVT, A fib, VA)
- ↓ occurrence of sudden cardiac death
- Slows the progression of CHF
- Has been known to ↓ mortality & morbidity in CHF patients

Beta-Adrenergic Blockers

- Decrease cardiac contractility and output
- ↓ CO by ↓ heart rate
- Diminishes the sympathetic reflex
- Decreases renin release
- Improves blood flow & heart rhythm problems
- ↓ myocardial oxygen demand in patients with Ischemic Heart Disease

Beta-Adrenergic Blockers

- Side effects include:
 - coldness, fatigue, mask hypoglycemia, depression, impotence, SOB, bradycardia
- Avoid abrupt withdrawal due to rebound HTN, cardiac disease and increase in the risk of MI or sudden death
- Side effects pronounced in the elderly
 - Taper over 1 to 2 weeks
- ↓ HDL & ↑ TGs: cautious use in hyperlipidemia

Beta-Adrenergic Blockers

- ▶ Caution should be used in **asthma** patients due to **bronchospasm**
- ▶ May mask insulin **induced hypoglycemia**
 - e.g., tremor, tachycardia, and palpitations: mediated through the sympathetic nervous system
 - Delay in hypoglycemia recovery time because: Inhibited Glycogenolysis
 - Unopposed α -receptor stimulation during the acute hypoglycemic recovery phase may result in acutely elevated BP

Central Alpha Agonists

- Clonidine: **works rapidly in hypertensive emergencies**
 - 0.1-0.8 mg BID
- Methyldopa: **used in pregnant women**
 - 250-1000 mg BID
- ▶ lower BP primarily by stimulating α_2 -adrenergic receptors in the brain
 - **Reduces sympathetic outflow from the vasomotor center in the brain and**
- ▶ Decrease heart rate, CO, TPR, plasma renin activity, and baroreceptor reflexes

Central Alpha Agonists

- ▶ Clonidine is often used in resistant hypertension
- ▶ methyldopa is a first-line agent for pregnancy-induced hypertension
- ▶ Chronic use can cause water/Na retention
 - Use with diuretics

Central Alpha Agonists

- ▶ Sudden withdrawal can cause rebound hypertension
 - Taper gradually
- ▶ Side effects includes:
 - orthostatic hypotension
 - rebound hypertension
 - dizziness, anxiety, agitation & confusion
 - Methyldopa: hepatitis

Direct Arterial Vasodilators

- ▶ Includes:
 - **Hydralazine: 20-100 mg BID or QID**
 - **minoxidil: 10-40 mg QD or BID**
 - Directly relax arteriolar smooth muscle resulting in vasodilation and BP lowering
- ▶ Results in activation of baroreceptors
 - compensatory increase in sympathetic outflow
 - leads to an increase in heart rate, CO, and renin release
 - loss of hypotensive effect with continued use

Direct Arterial Vasodilators

- ▶ Minoxidil is a more potent vasodilator than hydralazine
 - Can precipitate HF
 - co-administer a β -blocker and a diuretic
 - A loop diuretic is often more effective than a thiazide in patients treated with minoxidil

Direct Arterial Vasodilators

▶ side effect

▪ Hydralazine

- a dose-dependent drug-induced lupus-like syndrome (Vasculitis)
- reversible on discontinuation
- Less common when daily dose < 200 mg

▪ Minoxidil

▪ hypertrichosis (hirsutism)

- Increased hair growth on the face, arms, back, and chest
- reversible on discontinuation

Alpha Adrenergic Blockers

- ▶ Prazosin: 2-20 mg BID or TID
- ▶ Terazosin: 1-20 mg QD or BID
- ▶ Doxazosin: 1-8 mg QD

Alpha Adrenergic Blockers

- Causes relaxation of blood vessels
- Cross BBB and can cause CNS side effects
- **Dizziness** is a major complaint
- Works peripherally to increase vasodilatation
- In the first 3 hrs after the initial dose, patients may experience faintness, dizziness, palpitation, or syncope
- Avoid in elderly patients due to risk of falls

Alpha Adrenergic Blockers

- ▶ ↓ PVR by relaxing arterial & venous smooth muscles
- ▶ Common adverse effect: orthostatic hypotension and it can persist with chronic use
- ▶ Used in patients with benign prostatic hyperplasia (Terazosin)

Alpha Adrenergic Blockers

- Do not use as monotherapy
 - not as effective as other agents
- Due to fluid/Na retention with prolonged use and higher doses
 - Give concomitantly with diuretics
- Should be taken at bedtime to reduce syncope

Treatment Of Hypertension In Special Population

Hypertension in Older People

- Presents as **isolated systolic hypertension**
- **CV morbidity and mortality: directly correlated to SBP than to DBP for patients aged 50 and older**
- At high risk for hypertension related target-organ damage
- **Preferred agents: thiazide diuretics, ACE inhibitors, CCBs and ARBs**
 - Use smaller-than-usual initial doses
- **Centrally acting agents and α 1-blockers should generally be avoided or used with caution**
 - associated with dizziness and orthostatic hypotension

Treatment Of Hypertension In Special Population

Patients at Risk for Orthostatic Hypotension

- ▶ *Orthostatic hypotension*: a significant drop in BP when standing
 - A SBP decrease of >20 mm Hg or DBP decrease of >10 mm Hg when changing from supine to standing
- ▶ Associated with dizziness and/or fainting
- ▶ Risk of orthostatic hypotension increased in
 - Older patients, Diabetes, Severe volume depletion
 - baroreflex dysfunction, Autonomic insufficiency
 - Concomitant venodilators (α -blockers, mixed α -/ β -blockers, nitrates, and phosphodiesterase inhibitors)
- ▶ Start antihypertensive agents at low doses
 - ▶ Especially diuretics, ACE inhibitors, and ARBs

Treatment Of Hypertension In Special Population

Pregnancy

- ▶ **Methyldopa**: considered the drug of choice
- ▶ Reasonable alternatives
 - ▶ CCBs ...Nifedipine
 - ▶ Labetalol
 - ▶ *Other* β -Blockers (other than atenolol)
- ▶ ACE inhibitors and ARBs
 - Known teratogens
 - **Absolutely contraindicated**
- when DBP >105 mm Hg
 - IV hydralazine or IV labetalol

Antihypertensive Therapy In Concomitant Conditions

Pulmonary Disease and Peripheral Arterial Disease (PAD)

► Reactive airway disease (asthma or COPD)

- *Avoid β -Blockers*, especially nonselective agents
- fear of inducing bronchospasm
- Use cardioselective β -blockers

► PAD

- β -Blockers: problematic
- possible decrease in peripheral blood flow
- secondary to unopposed stimulation of α 1-receptors
- *Use β -blocker that also has α 1-blocking properties (e.g., carvedilol)*

Antihypertensive Therapy In Concomitant Conditions

Metabolic Syndrome

- ▶ defined as the presence of three of the following five criteria:
 - Abdominal obesity (based on waist circumference measurements)
 - Elevated triglycerides
 - Low HDL cholesterol
 - Elevated BP ($\geq 130/\geq 80$ mm Hg or receiving drug treatment for high BP), and
 - Elevated fasting blood glucose
- ▶ Increased risk of developing CV disease
- ▶ **Use an ACE inhibitor or ARB**

Antihypertensive Therapy In Concomitant Conditions

Erectile Dysfunction

- ▶ Associated with most antihypertensive agents
- ▶ β -Blockers mostly associated with ED
- ▶ Avoid centrally acting agents
- ▶ Atherosclerotic vascular disease may contribute to ED
 - Optimal control of BP

Resistant Hypertension

- ▶ Defined as hypertension which is uncontrolled
 - failure to achieve goal BP of $<130/80$ mm Hg, or lower when indicated with the use of three or more drugs on maximum dose that includes a diuretic
 - Ideally in patients who are adhering to full doses of an appropriate three-drug regimen that includes a diuretic
- ▶ Includes patients who are controlled but require the use of four or more medications

Resistant Hypertension

▶ Refractory hypertension

▶ failure to control BP despite use of at least 5 antihypertensive agents of different classes, including

- A long acting thiazide-type diuretic, such as chlorthalidone, and
- A mineralocorticoid receptor antagonist, such as spironolactone

Resistant Hypertension

- ▶ Caused by
 - ▶ Improper BP measurement
 - ▶ Volume overload
 - Excess sodium intake
 - Kidney disease
 - Inadequate diuretic therapy
 - ▶ Drug induced
 - ▶ Non adherence
 - ▶ Obesity
- ▶ Excess alcohol intake

Resistant Hypertension

Treatment

- assuring adequate diuretic therapy
 - Adding an aldosterone antagonist
- appropriate use of combination therapies
- using alternative antihypertensive agents when needed

Hypertensive Urgency

- Gradual reduction in BP is preferred
- Managed by adjusting maintenance therapy
 - By adding a new antihypertensive, and/or by increasing the dose of a present medication
 - BP reductions with oral antihypertensive agents to stage 1 values over a period of several hours to several days
 - Reevaluate patients within and no later than 7 days (preferably after 1 to 3 days)

Hypertensive Emergency

- ▶ Require immediate BP reduction
 - To limit new or progressing target-organ damage
- ▶ Require parenteral therapy
- ▶ Initial target is a reduction in MAP of up to 25% within minutes to hours
 - If the patient is stable, to 160/100 mm Hg within the next 2 to 6 hours
- ▶ Precipitous drops in BP may lead to end organ ischemia or infarction
- ▶ Gradual reductions toward goal BP values can be attempted after 24 to 48 hours
 - Except in acute ischemic stroke where maintaining an elevated BP is needed for a longer period of time

Drugs for hypertensive emergency

Nitroprusside

- ▶ Potent vasodilator
- ▶ Direct vasodilatation on smooth muscle to decrease arterial pressure
- ▶ Usual rate of administration is 0.3-10mcg/kg/min
- ▶ S/E: reflex tachycardia, hypotension

Drugs for hypertensive emergency

Labetalol

- ▶ Alpha & Beta blocker
- ▶ Fast onset and long duration of action
- ▶ Infusion is 0.5-2mg/min
- ▶ No chronotropic or inotropic effect on the heart
- ▶ Does not cause tachycardia
- ▶ Venous and arterial dilator & relaxes vascular smooth muscle
- ▶ Contraindicated in bronchial asthma pts & bradycardic patients
- ▶ Causes marked postural hypotension in intracranial hemorrhage patients

Drugs for hypertensive emergency

Nitroglycerin

- ▶ Dilates both arterial & venous vessels
- ▶ Decreases preload & afterload
- ▶ Increases blood supply to ischemic cardiac regions
- ▶ Usual dose is 5-300mcg/min
 - ▶ Patients may develop nitrate tolerance
 - provide nitrate free intervals

Drugs for hypertensive emergency

Hydralazine

- ▶ Vasodilator that ↓ peripheral vascular resistance by direct action on the vascular smooth muscle
- ▶ Usual dose is 5-40mg IV/IM Q4-6H
- ▶ Do not exceed 5mg/min IV push
- ▶ May cause lupus, reflex tachycardia, dizziness, palpitations, joint pains
- ▶ Contraindicated in patients with aortic aneurysm
- ▶ Also used in pregnancy

Monitoring

- ▶ Laboratory monitoring should typically occur 2 to 4 weeks after starting a new agent or dose increase
 - Every 6 to 12 months in stable patients
 - Eplerenone or spironolactone: potassium concentrations and kidney function within 3 days of initiation and again at 1 week to detect potential hyperkalemia
 - Adherence with each office visit
 - At monthly intervals until control is achieved
- ▶ For adults with a normal BP, repeat evaluation every year is reasonable

Monitoring

Class	Parameters
Aldosterone antagonists	Blood pressure; BUN/serum creatinine; serum potassium
ACE inhibitors	Blood pressure; BUN/serum creatinine; serum potassium
ARBs	Blood pressure; BUN/serum creatinine; serum potassium
Calcium channel blockers	Blood pressure; heart rate
Diuretics	Blood pressure; BUN/serum creatinine; serum electrolytes (potassium, magnesium, sodium); uric acid (for thiazides)
β -Blockers	Blood pressure; heart rate